

Natamycin Blocks Fungal Growth by Binding Specifically to Ergosterol without Permeabilizing the Membrane^{*§}

Received for publication, September 18, 2007, and in revised form, November 30, 2007 Published, JBC Papers in Press, December 29, 2007, DOI 10.1074/jbc.M707821200

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Natamycin is a polyene antibiotic that is commonly used as an antifungal agent because of its broad spectrum of activity and the lack of development of resistance. Other polyene antibiotics, like nystatin and filipin are known to interact with sterols, with some specificity for ergosterol thereby causing leakage of essential components and cell death. The mode of action of natamycin is unknown and is investigated in this study using different *in vitro* and *in vivo* approaches. Isothermal titration calorimetry and direct binding studies revealed that natamycin binds specifically to ergosterol present in model membranes. Yeast sterol biosynthetic mutants revealed the importance of the double bonds in the B-ring of ergosterol for the natamycin-ergosterol interaction and the consecutive block of fungal growth. Surprisingly, in strong contrast to nystatin and filipin, natamycin did not change the permeability of the yeast plasma membrane under conditions that growth was blocked. Also, in ergosterol containing model membranes, natamycin did not cause a change in bilayer permeability. This demonstrates that natamycin acts via a novel mode of action and blocks fungal growth by binding specifically to ergosterol.

Fungal infections have recently become a growing threat to human health, especially in persons whose immune systems are compromised (for example, by human immunodeficiency virus and cancer chemotherapy). Only a few effective antifungal agents are currently in use; these include the polyenes, the fluorocytosines, and the azole derivatives. One important problem is the increase of drug resistance, particularly against azole antimycotics and fluorocytosine (1). Resistance against polyene antibiotics is still a rare event, which makes these antibiotics particularly interesting as antifungal agents. The polyene antibiotics have a ring structure in which a conjugated double bond system is located opposite to a number of hydroxyl functions. Often a mycosamine group is present in combination with a carboxyl moiety, rendering the molecule amphoteric (Fig. 1). In the past convincing evidence has been presented that several members of this class of antibiotics target sterols and in particular ergosterol, the abundant and main sterol of fungal membranes (2, 3). Different types of polyene antibiotics were shown to have different modes of action despite that they share a common target. The larger polyenes like amphotericin B and nystatin form pores together with ergosterol in the plasma membrane that collapse vital ion gradients, thereby killing the cells. The smaller uncharged filipin also destroys the membrane barrier, but by a completely different mechanism. Filipin forms large complexes with sterols between the leaflets of the lipid bilayer, resulting in loss of the barrier function (2). Natamycin (also called pimarinic acid) is a very effective member of the polyene antibiotic family with a large standing record of applications. It is produced by *Streptomyces nateus* and used against fungal infections, but it is also widely utilized in the food industry to prevent mold contamination of cheese and other nonsterile foods (e.g. cured meats) (4). Surprisingly, the mechanism of action of this antifungal agent is still unknown and it is even unknown whether it targets ergosterol in the fungal membrane. It is relatively small while it contains a tetraene compared with a pentaene in filipin, which is already considered as a small polyene antibiotic (Fig. 1). It contains a mycosamine group that renders it amphoteric, which is a feature that is also present in nystatin. Whereas natamycin has similar features of both filipin (small) and nystatin (amphoteric), it is difficult to predict its mechanism of action.

We wanted to gain more insight into the mode of action of natamycin, which could in turn help to develop new or improved antifungal formulations or result in novel strategies to prevent fungal spoilage. To determine the interaction of natamycin with membranes in relation to its sterol composition, we tested in a comparative manner using filipin and nystatin as references, the interaction of natamycin with phosphatidylcholine model membranes of varying sterol composition using isothermal titration calorimetry (ITC)² and other binding studies. In addition, the ability of natamycin to permeabilize these model membranes was studied.

Parallel to the studies performed on model membranes, the effect of natamycin on yeast growth, the binding of the anti-

^{*} This work was supported by the Technology Foundation (STW), Applied Science division of the Netherlands Organization for Scientific Research (NWO) (UBC.6524). The costs of publication of this article were defrayed in part by the payment of page charges. This article must therefore be hereby marked "advertisement" in accordance with 18 U.S.C. Section 1734 solely to indicate this fact.

[§] The on-line version of this article (available at <http://www.jbc.org>) contains supplemental Fig. S1.

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² The abbreviations used are: ITC, isothermal titration calorimetry; DOPC, 1,3-dioleoyl-sn-glycero-3-phosphocholine; CFDA-SE, 5-(and -6)-carboxyfluorescein diacetate, succinimidyl ester; HPTS, 8-hydroxypyrene-1,3,6-trisulfonic acid trisodium salt; DDAO, N,N-dimethyldecylamine-N-oxide; MIC, minimum inhibitory concentration; CF, carboxyfluorescein; LUVs, large unilamellar vesicles; MES, 4-morpholineethanesulfonic acid.

otic with intact yeast cells, and the plasma membrane integrity were determined. These studies were performed using strains that carry specific mutations in the ergosterol biosynthetic pathway (*ergΔ*) or that were reprogrammed to contain cholesterol as the main sterol (5). We could demonstrate that, differently from any other polyene antibiotic of which the mode of action is known, natamycin blocks fungal growth by binding specifically to ergosterol, but without permeabilizing the membrane.

EXPERIMENTAL PROCEDURES

Chemicals—1,2-Dioleoyl-*sn*-glycero-3-phosphocholine (DOPC) and cholesterol were purchased from Avanti Polar Lipids Inc. (Alabaster, AL). Ergosterol was purchased from Larodan AB (Sweden). DOPC or sterols were dissolved in chloroform to a stock concentration of 20 mM. The phospholipid concentration of DOPC was determined by phosphate analysis according to Rouser *et al.* (6). The polyene antibiotics nystatin and filipin were dissolved in Me₂SO, whereas natamycin was dissolved in 85:15 Me₂SO/H₂O (v/v); all were obtained from Sigma. All antibiotic solutions were prepared freshly before the start of an experiment and the concentrations of the polyene antibiotics were determined by UV absorption on a PerkinElmer UV-visible spectrometer (Lambda 18). The molar extinction coefficients of the polyene antibiotics were determined in methanol to be $7.6 \times 10^4 \text{ M}^{-1} \text{ cm}^{-1}$ (318 nm), $6.7 \times 10^4 \text{ M}^{-1} \text{ cm}^{-1}$ (318 nm), and $8.5 \times 10^4 \text{ M}^{-1} \text{ cm}^{-1}$ (356 nm) for natamycin, nystatin, and filipin, respectively. The molar extinction coefficient of ergosterol was measured in methanol to be $0.97 \times 10^4 \text{ M}^{-1} \text{ cm}^{-1}$ (262 nm).

The ionophore nigericin (dissolved in ethanol), ampicillin sodium salt, and the amino acids adenine, uracil, and L-tryptophan were obtained from Sigma. 5-(and -6)-Carboxyfluores-

cein diacetate, succinimidyl ester (CFDA-SE) (dissolved in Me₂SO) and 8-hydroxypyrene-1,3,6-trisulfonic acid trisodium salt (HPTS) were both purchased from Invitrogen. *N,N*-Dimethyldodecylamine-*N*-oxide (DDAO) was bought from Fluka Biochimica (Buchs). All other chemicals used were of analytical or reagent grade.

Strains and Growth Conditions—For all experiments, medium was inoculated directly from plates with colonies that were not older than 2 weeks. Unless otherwise mentioned, cells were grown overnight at 30 °C in rich medium (10 g/liter yeast extract, 20 g/liter Bacto-peptone, and 20 g/liter dextrose with 1 g/liter adenine, 2 g/liter uracil, and 1 g/liter tryptophan (YPUADT)) supplemented with 0.1 mg/ml ampicillin. For strains RH6611 and RH6613 SD medium was used (1.7 g/liter yeast nitrogen base without amino acids, 20 g/liter glucose, 2 mg/liter trace components, 5 g/liter ammonium sulfate) supplemented with vitamins and the appropriate amino acids minus histidine and leucine (SD-His-Leu). Yeast strains used in this study are listed with their relevant genotypes in Table 1 and the plasmids in Table 2.

MIC Value Determinations—Minimum inhibitory concentrations (MICs) were determined by diluting the polyene antibiotics in YPUADT (with 0.1 mg/ml ampicillin) to concentrations of 400, 350, 300, and 250 μM of which 100 μl was added to the first row of a 96-well suspension culture plate (U-form, Greiner Bio One). This was followed by a 1:1 dilution series in medium. Overnight cultures were diluted back to an A₆₀₀ 0.0001, of which 100 μl was added to the culture plate. The total volume per well was 200 μl. Strains RH6611 and RH6613 (in SD-His-Leu medium) were diluted to an A₆₀₀ 0.01, because they had a very slow growth rate. The MIC value was determined to be the lowest concentration of antibiotic, which inhibits the growth of the yeast strain and could be determined by eye on the 96-well plate after an incubation of 24 h at 30 °C. The experiments were performed in triplicate.

Preparation of Large Unilamellar Vesicles (LUVs)—LUVs with a mean diameter of 200 nm were prepared using the following protocol. Aqueous phospholipid suspensions were prepared by premixing ergosterol or cholesterol with DOPC in the desired molar ratios as solutions in chloroform and evaporating the solvent in a stream of nitrogen, followed by drying the lipid film for 20 min under vacuum. Sterols were present in a range of 10 to 30 mol %. All following handlings were performed at 50 °C. The lipid film was hydrated and repeatedly vortexed until all lipid was removed from the walls of the test tube. Then a freeze-thaw cycle was repeated eight times using liquid nitro-

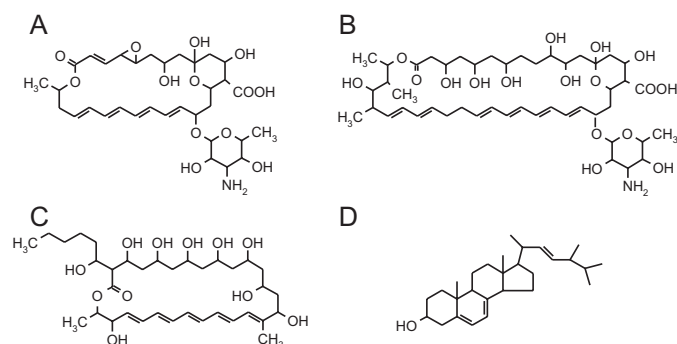


FIGURE 1. Structures of several polyene antibiotics and ergosterol. A, natamycin; B, nystatin; C, filipin; D, ergosterol.

TABLE 1

Strains used in this study

The source of these strains is described in Ref. 5.

Strain	Name	Genotype
Wild type	RH448	<i>MATa his4 leu2 ura3 lys2 bar1</i>
Erg2Δ	RH2897	<i>MATa erg2(end11)-1Δ::URA3 leu2 ura3 his4 lys2 bar1</i>
Erg2Δerg6Δ	RH3616	<i>MATa erg2(end11)-1Δ::URA3 erg6Δ leu2 ura3 bar1</i>
Erg6Δ	RH3622	<i>MATa erg6Δ::LEU2 leu2 ura3 his4 bar1</i>
Erg3Δ	RH4213	<i>MATa erg3Δ::LEU2 leu2 ura3 his4 lys2 bar1</i>
Erg3Δerg6Δ	RH5225	<i>MATa erg3Δ::LEU2 erg6Δ::LEU2 leu2 ura3 his4 lys2 bar1</i>
Erg2Δerg3Δ	RH5228	<i>MATa erg2Δ (end11)-1Δ::URA3 erg3Δ::LEU2 leu2 ura3 his4 lys2 bar1</i>
Erg4Δerg5Δ	RH5233	<i>MATa erg4Δ::URA3 erg5Δ::kanMX4 leu2 ura3 his4 lys2 bar1</i>
Wild type	RH6611	<i>MATa his3 ura3 leu2 (pRS423) (pRS425)</i>
Cholesterol	RH6613	<i>MATa erg5Δ::TRP1 erg6::TRP1 his3 ura3 leu2 trp1 (pRS423-DHCR7) (pRS425-DHCR24)</i>

TABLE 2
Plasmids used in this study

Plasmid	Characteristics	Ref.
pRS423	Multicopy vector containing <i>GDP</i> promoter and <i>HIS3</i>	30
pRS423-DHCR7	pRS423 derivative vector containing <i>DHCR7</i> gene	— ^a
pRS425	Multicopy vector containing <i>GDP</i> promoter and <i>LEU2</i>	30
pRS425-DHCR24	pRS425 derivative vector containing <i>DHCR24</i> gene	— ^a

^a C. M. Souza, H. Pichler, E. Leitner, X. Guan, M. R. Wenk, I. Tornare, and H. Riezman, submitted for publication.

gen and a water bath. Subsequently, the lipid suspension was extruded 8 times through a polycarbonate membrane filter with a pore size of 0.2 μm (Whatman International). The size of the vesicles was determined after extrusion by using the Zeta-sizer 3000 (Malvern Instruments). The average of the size of the vesicles was 168 ± 3.7 nm for vesicles without sterols, 165 ± 1.2 nm for vesicles with 10% cholesterol, and 173 ± 8 nm for vesicles with 10% ergosterol. Thus no significant differences in size were observed. The resulting vesicle suspension was stored at 4 °C. The final phospholipid concentration was determined by phosphate analysis according to Rouser *et al.* (6).

ITC Measurements—Titration experiments were carried out on a MCS titration calorimeter from Microcal Inc. LUVs were prepared as described above in 50 mM MES, 100 mM K_2SO_4 , pH 6.0, or 10 mM HEPES, 100 mM NaCl, pH 7.0. Similar results were obtained with the different buffers. The vesicles were injected into a sample cell (volume = 1.345 ml) containing 50 μM antibiotic in the same buffer as used for the vesicle suspension. Because the polyene antibiotics are dissolved in Me_2SO , an equal amount was added to the LUV suspension to compensate for any heat generated by dilution of this solvent. No more than 1% Me_2SO was present. The solutions were degassed, before the start of the titration. The experiments consisted of 44 injections, 5 μl each, of a stock solution of vesicles at 25 °C (8 mM final phospholipid concentration). The results were analyzed using the ORIGIN software (version 2.9) provided by Microcal Inc. The interaction between the vesicles and the antibiotics was complex in that no clear saturation of this interaction was observed. Therefore the stoichiometry of the interaction could not be determined. An approximation of the binding constant was made using the ORIGIN software, where the value of integrated heat of the last injection was subtracted from all data and the model of one set of sites was fitted to the resulting data.

Binding Assay Using Centrifugation of Model Membranes—Vesicles were prepared as described above in 10 mM MES/Tris, 15 mM K_2SO_4 at pH 7. The reduced ion strength facilitated the pelleting of the vesicles. The concentrations of antibiotics and vesicles were varied from 0 to 0.1 and 0.5 to 5 mM, respectively, unless indicated otherwise. Vesicles were incubated with the polyene antibiotics for 1 h in an Eppendorf incubator (22 °C, 650 rpm), with a maximum of 1% Me_2SO present. To spin down the vesicles and the bound antibiotic, 1 ml of the mixture was centrifuged in a TLA 120.2 rotor in a Beckman Ultracentrifuge (TL-100) for 1.5 h at 100 krpm and 20 °C. The amount of antibiotic before centrifugation and in the supernatant and pellet was determined by UV absorption after 7 times dilution in methanol followed by centrifugation to remove any precipi-

tated salts. The phospholipid concentrations were determined by phosphate analysis according to Rouser *et al.* (6). Under these conditions less than 10% of the phospholipids remained in the supernatant. The antibiotics were not pelleted in the absence of lipid below a concentration of 75, 34, and 30 μM , respectively, of natamycin, nystatin, and filipin. The binding isotherms of the interaction of natamycin with ergosterol could be described by the Langmuir adsorption model assuming that ergosterol was the only binding site for natamycin in the DOPC vesicles and that only the ergosterol in the outer leaflet of the bilayer could have an interaction with natamycin. The Langmuir adsorption model was applied to the data of the amount of natamycin bound to the vesicles *versus* the amount of free natamycin in the supernatant (7). From using this model in Sigmaplot (10.0), the binding constant and the binding saturation of natamycin with ergosterol could be determined.

Binding Assay Using Centrifugation of Intact Cells—Yeast were grown to the mid-logarithmic phase in 200 ml of YPUADT (with 0.1 mg/ml ampicillin) or SD medium. As a negative control the *Escherichia coli* strain DH5 α was used that was grown to the logarithmic phase in 100 ml of Luria Broth (LB) medium at 37 °C. The cells were harvested by centrifugation at room temperature at $3600 \times g$ for 10 min in a Sorvall RC 5B centrifuge (SLA 1500), washed two times in 100 ml of 10 mM MES/Tris, 15 mM K_2SO_4 at pH 7, and resuspended in a small volume of buffer. The A_{600} of the cell suspensions was determined and a series of 1-ml cell suspensions were prepared ranging from an A_{600} of 0 to 15. The cells were centrifuged at $3000 \times g$ for 5 min at room temperature and resuspended in the same buffer containing 30 μM natamycin. As a control, cells were resuspended in buffer with no natamycin. The cells were incubated for 1 h in an Eppendorf incubator (900 rpm at room temperature) and spun down for 15 min at $3000 \times g$. The amount of natamycin in the supernatant was determined by UV absorption as described above (spectrum from 250 to 350 nm) and used to calculate the amount of natamycin bound to the yeast cells.

Carboxyfluorescein Permeability Assay in Large Unilamellar Vesicles—Carboxyfluorescein (CF)-loaded vesicles were prepared as described above in 50 mM MES/KOH buffer at pH 7 (8). To remove the untrapped CF, a Sephadex G-50 spin column equilibrated with 50 mM MES, 100 mM K_2SO_4 buffer at pH 7 was used. The CF-loaded vesicles were diluted in 1200 μl of 50 mM MES, 100 mM K_2SO_4 buffer at pH 7 followed by the addition of the antibiotic. The antibiotic-induced CF leakage from the vesicles was monitored by measuring the fluorescence intensity at 513 nm (excitation set at 430 nm) on a SLM AMINCO Spectrofluorometer (SPF-500). The detergent Triton X-100 was added at the end of the experiment to destroy the lipid vesicles and the resulting fluorescence was taken as the 100% leakage value.

Proton Permeability Assay in Large Unilamellar Vesicles—Proton permeability was determined in an assay with HPTS-loaded vesicles as performed by van Kan *et al.* (9). The assay is based on the strong pH dependence of the fluorescence of HPTS. Vesicles were prepared as described above in a 2 mM HPTS solution in 0.2 M $\text{NaH}_2\text{PO}_4/\text{Na}_2\text{HPO}_4$ buffer at pH 7. To create a lower pH at the outside and remove all the untrapped

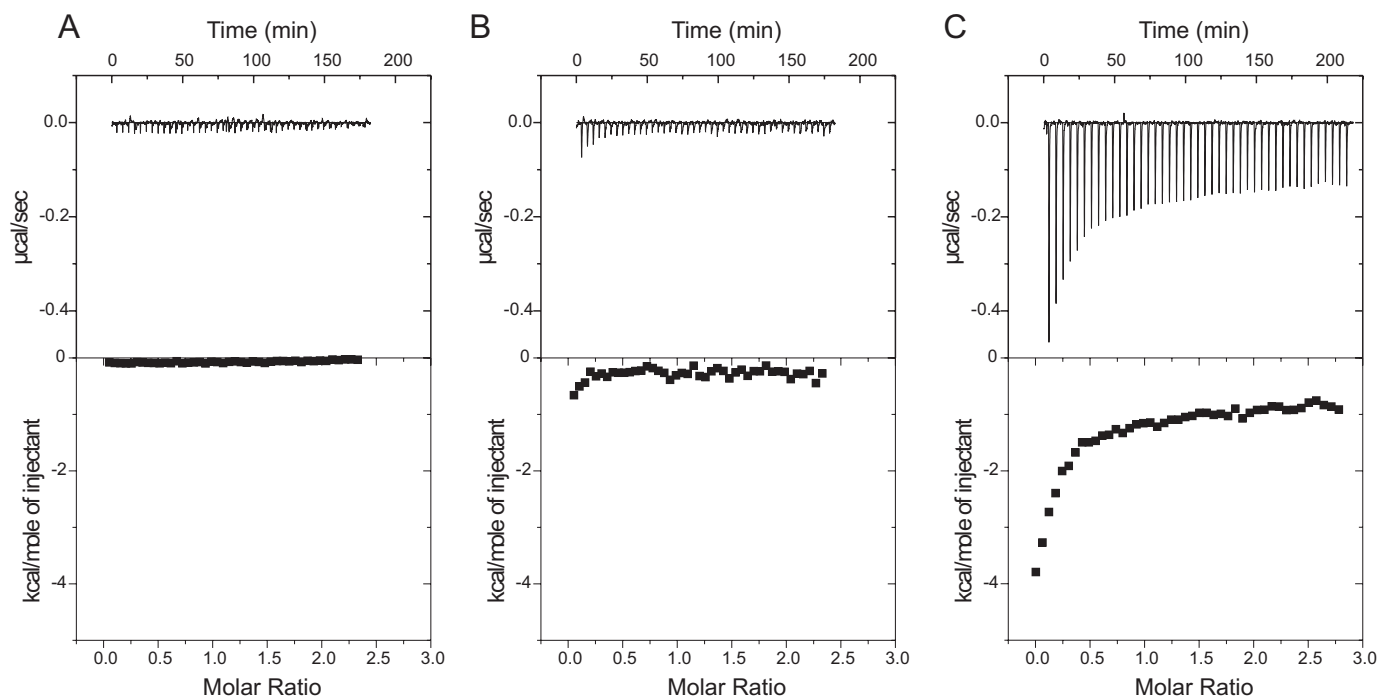


FIGURE 2. **Calorimetric titrations of natamycin with DOPC vesicles.** Vesicles contained no sterol (A), 10% cholesterol (B), and 10% ergosterol (C) and were dissolved in 50 mM MES, 100 mM K_2SO_4 , pH 6.0. The *top graph* displays the heat peaks after consecutive injections of 5- μ l vesicles with an 8 mM final phospholipid concentration into the sample cell containing 50 μ M natamycin. The *bottom graph* shows the integrated heat per injection, which is normalized to the injected amount of moles of sterol and is displayed against the molar ratio of sterol *versus* natamycin. When no sterols are present, 10% of phospholipid is used to determine and display the integrated heat per injection.

HPTS, a Sephadex G-25 spin column was used equilibrated with 10 mM MES, 0.2 M Na_2SO_4 buffer at pH 5.5. To determine the phospholipid concentration of the resulting vesicles the lipids were first extracted according to Bligh-Dyer (10) to exclude the phosphate from the buffer in the following phosphate analysis according to Rouser *et al.* (6). The effects of the polyene antibiotics on the proton permeability of the lipid vesicles was monitored by adding aliquots of antibiotic to 1200 μ l of 10 mM MES and 0.2 M Na_2SO_4 buffer, pH 5.5, containing HPTS-loaded vesicles (35 μ M phospholipid phosphorous). The fluorescence emission was detected at 508 nm (excitation at 450 nm) on a SLM AMINCO Spectrofluorometer (SPF-500). Differing from van Kan *et al.* (9), the detergent DDAO was used instead of Triton X-100, because DDAO did not have any effect on the fluorescence of the probe where Triton X-100 did have an effect (not shown). DDAO was added at the end to destroy the lipid vesicles and the resulting fluorescence was taken as the 100% leakage value, whereas the blank without antibiotic was used as a reference for 0% leakage. Nigericin, a polyether ionophore known to collapse proton gradients, was used as a positive control (11).

Proton Permeability Assay in Yeast—The assay was based on the loading of yeast cells with the probe CFDA-SE as described by Bracey *et al.* (12, 13). CFDA-SE is a non-polar molecule that spontaneously penetrates cell membranes and is converted to the anionic pH-sensitive 5-(and-6)-carboxyfluorescein succinimidyl ester (CF-SE) by intracellular esterases (9). Once the probe is internalized, amine reactive coupling of succinimidyl groups of CF-SE to aliphatic amines of intracellular proteins results in the formation of membrane-impermeable pH-sensitive probe conjugates.

Wild type yeast cells from an overnight culture were diluted to an A_{600} of ~ 0.8 and then centrifuged at $3000 \times g$ for 3 min. The cells were washed and resuspended in an equal volume of 100 mM citric/phosphate buffer at pH 4 (100 mM citric acid, 50 mM NaH_2PO_4 , and 50 mM KOH). CFDA-SE (100 μ M) was added and the cells were incubated overnight while shaking at 37 $^{\circ}C$. The viability of the cells was not significantly compromised by the loading conditions. Loaded cells were harvested ($3000 \times g$ for 3 min), washed, and resuspended in YPUADT buffered with 50 mM citric/phosphate (pH 4) to an A_{600} of 0.4. To recover from the stress imposed by the probe loading conditions, the cultures were left for 1 h at 30 $^{\circ}C$ with shaking. The effects of the polyene antibiotics on the proton permeability of the yeast cells were monitored by adding aliquots of antibiotic to 5 ml of culture and measuring the A_{600} and fluorescence at regular intervals. The A_{600} was determined on a Helios Epsilon UNICAM spectrometer and the fluorescence emission was detected at 525 nm (excitation at 495 nm) on a SLM AMINCO Spectrofluorometer (SPF-500).

RESULTS

Sterol Specificity of Natamycin Binding to Membranes—To test whether sterols are required for membrane affinity of natamycin we used phosphatidylcholine model membranes containing ergosterol, the main fungal sterol or cholesterol, the main sterol in mammals.

The interaction between natamycin and sterols in the model membrane was first studied using ITC. ITC measurements were performed where LUVs containing either no sterols, cholesterol, or ergosterol were titrated into a solution of natamycin (Fig. 2). Natamycin displayed no interaction with vesicles containing no sterols as the resulting heats were no different from

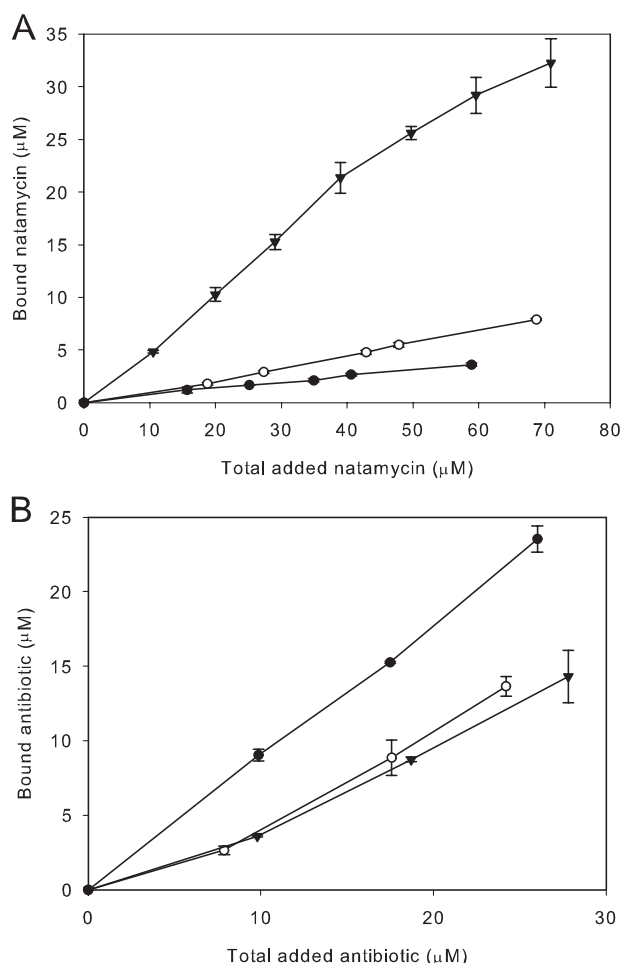


FIGURE 3. Interaction of polyene antibiotics with model membranes. A, binding of natamycin to vesicles containing 10% ergosterol (▼), 10% cholesterol (○), or no sterols (●). B, the interaction of filipin (●), nystatin (○), and natamycin (▼) on 10% ergosterol containing vesicles was examined. The assay was performed in duplicate in 10 mM MES/Tris, 15 mM K_2SO_4 , pH 7.0, and the vesicles had a 2 mM final phospholipid concentration.

the control (Fig. 2A). LUVs containing 10 mol % cholesterol produced only minor heat effects during the first injections, which indicates that natamycin displayed only a very small interaction with cholesterol containing vesicles (Fig. 2B). Interestingly, 10 mol % ergosterol containing vesicles displayed a significant amount of interaction with natamycin as evidenced by the consecutive heat effects (Fig. 2C). This titration curve differs from a normal titration curve as no clear saturation of the interaction was observed. The binding constant between natamycin and ergosterol was estimated to be $5.7 \times 10^4 \text{ M}^{-1}$ (see "Experimental Procedures"). Comparable large differences in effects between cholesterol and ergosterol were observed for sterol concentrations of 20 mol % (data not shown).

Furthermore, the binding of natamycin to vesicles was studied by separating the bound from the free natamycin by centrifugation. Fig. 3A shows a representative graph of these results, from which can be concluded that ergosterol containing vesicles had a significant interaction with natamycin. In the absence of sterols or in the presence of cholesterol very little interaction with natamycin was observed consistent with the ITC experiments (Fig. 2). A similar sterol dependence of natamycin binding was observed when varying the concentrations of vesicles

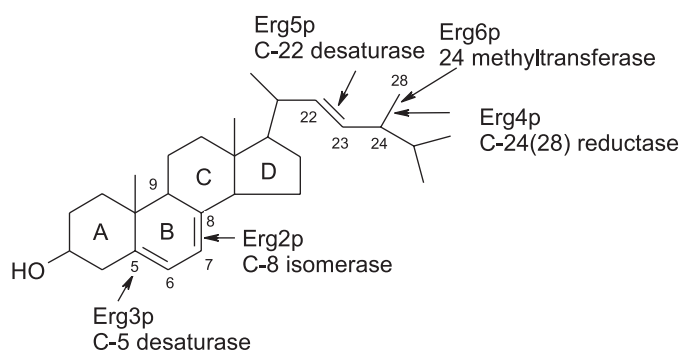


FIGURE 4. Ergosterol molecule with the assignment of the ring structure. Erg proteins and their functions are indicated. The corresponding genes are inactivated in *ergΔ* strains (5).

(data not shown). The binding constant was determined by the Langmuir adsorption model in SigmaPlot (10.0) to be $2.5 \pm 1.0 \times 10^4 \text{ M}^{-1}$, which is in reasonable agreement with the binding constant determined in the ITC measurements. The binding saturation from the Langmuir adsorption model was determined at $72 \pm 12 \mu\text{M}$ by extrapolating the data in SigmaPlot (10.0). By assuming that only the sterol in the external leaflet of the lipid vesicles could establish an interaction with the antibiotic, the sterol to antibiotic ratio was calculated to be $\sim 1:1$. If all sterols would be available for the interaction, because of sterol flip-flop, the ratio would be 2:1.

The affinity of natamycin for ergosterol containing vesicles was compared with that of filipin and nystatin to get insight into the relative strength of this interaction. Fig. 3B shows a representative graph of the results obtained with these antibiotics. Of the three polyene antibiotics filipin showed the highest affinity, followed by natamycin and nystatin.

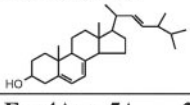
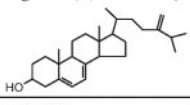
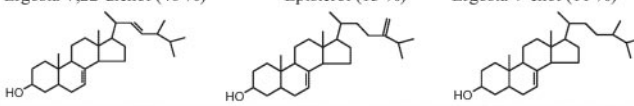
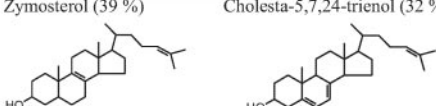
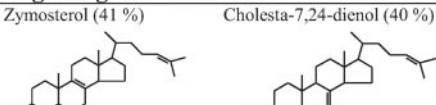
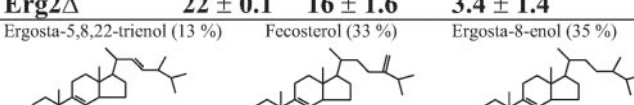
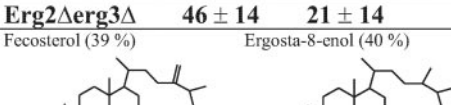
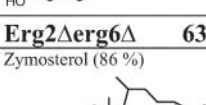
Sterol Specificity in the Antibiotic Action—To test if ergosterol is needed for natamycin to exert its antifungal activity *in vivo*, yeast strains carrying specific mutations in the ergosterol biosynthesis pathway (*ergΔ*) were used. Because of these mutations, the strains cannot synthesize ergosterol. However, they each accumulate a distinct set of sterols that, compared with ergosterol, have structural differences in the side chain and double bonds in the B or C ring (Fig. 4). The availability of these strains allows us to address the sterol specificity for polyenes, in relation to their inhibitory activity.

The most prominent sterols present in the *ergΔ* mutants are tabulated in percentage of total sterol present, together with their MIC values for the polyene antibiotics natamycin, nystatin, and filipin in Table 3. The sterol composition of the *erg* strains given in Table 3 was taken from Heese-Peck *et al.* (5) and specifies the percentage of a listed sterol compared with the total sterol composition of a cell. The most sensitive *erg* strain is *erg4Δerg5Δ*, which has a MIC value of the wild type strain. The least sensitive toward natamycin was *erg2Δerg6Δ*, which contained mostly zymosterol. From the strain with the highest sensitivity toward the lowest, the most striking sterol structural feature that causes the loss of activity is the loss of double bonds in ring B. For example, the sterols in *erg3Δ* have one double bond at position C-7,8 and it is only 3 times less sensitive to natamycin compared with the wild type, whereas *erg2Δerg6Δ* has lost both double bonds at C-5,6 and C-7,8 and is 37 times

TABLE 3

The minimum concentration of the polyene antibiotics needed to inhibit the growth of different *ergΔ* mutants

The MIC values for natamycin (MIC_{natam}), nystatin (MIC_{nyst}), and filipin (MIC_{filip}) are given for the different *ergΔ* strains, together with the structure and percentage of the most abundant sterols in an *ergΔ* strain, as stated in Ref. 5. The MIC values were determined in triplicate.

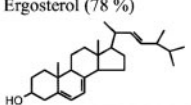
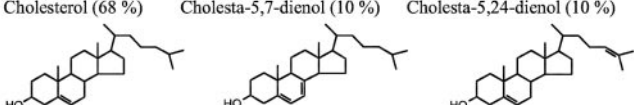
Strain	MIC _{natam} (μM)	MIC _{nyst} (μM)	MIC _{filip} (μM)
Wild Type	1.7 ± 0.5	1.1 ± 0.2	1.2 ± 0.1
Ergosterol (77 %)			
Erg4Δerg5Δ	2.1 ± 0.6	2.1 ± 0.2	2.5 ± 0.9
Ergosta-5,7,24-trienol (72 %)			
Erg3Δ	5.7 ± 0.8	3.4 ± 0.5	2.5 ± 0.2
Ergosta-7,22-dienol (46 %)	Episterol (13 %)	Ergosta-7-enol (11 %)	
			
Erg6Δ	8.3 ± 0.9	7.8 ± 1.6	2.9 ± 0.2
Zymosterol (39 %)	Cholesta-5,7,24-trienol (32 %)		
			
Erg3Δerg6Δ	18 ± 3.6	14 ± 4.1	5.2 ± 0.9
Zymosterol (41 %)	Cholesta-7,24-dienol (40 %)		
			
Erg2Δ	22 ± 0.1	16 ± 1.6	3.4 ± 1.4
Ergosta-5,8,22-trienol (13 %)	Fecosterol (33 %)	Ergosta-8-enol (35 %)	
			
Erg2Δerg3Δ	46 ± 14	21 ± 14	6.8 ± 2.3
Fecosterol (39 %)	Ergosta-8-enol (40 %)		
			
Erg2Δerg6Δ	63 ± 0.1	52 ± 13	6.8 ± 0.9
Zymosterol (86 %)			

less sensitive compared with the wild type. Variations in the C₁₇ side chain of the sterols did not have very large effects on the sensitivity toward natamycin, which can be observed when comparing *erg4Δerg5Δ* with the wild type. The yeast strain sensitivities toward nystatin were similar compared with natamycin. Filipin sensitivity seemed not to be so dependent on the sterol structure. The results demonstrate that double bonds in the B ring of the sterols are very important for natamycin to

TABLE 4

The minimum concentration of the polyene antibiotics needed to inhibit the growth of strains RH6611 and 6613

The MIC values for natamycin (MIC_{natam}), nystatin (MIC_{nyst}), and filipin (MIC_{filip}) are given for the different strains, together with the sterol structure and percentage of the most abundant sterols in the strain, as stated (C. M. Souza, H. Pichler, E. Leitner, X. Guan, M. R. Wenk, I. Tornare, and H. Riezman, submitted for publication). The MIC values were determined in triplicate.

	MIC _{natam} (μM)	MIC _{nyst} (μM)	MIC _{filip} (μM)
Wild Type	2.0 ± 0.4	0.9 ± 0.1	2.7 ± 0.4
Ergosterol (78 %)			
Cholesterol	31 ± 6	13 ± 2.4	3.3 ± 0.6
Cholesterol (68 %)	Cholesta-5,7-dienol (10 %)	Cholesta-5,24-dienol (10 %)	
			

inhibit the growth of yeast, whereas changes of the C₁₇ side chain are of less importance.

Recently a yeast strain was constructed (RH6613) that is unable to synthesize ergosterol or its related precursors, but instead was programmed to synthesize cholesterol. This enabled us to test the strong preference of natamycin for ergosterol over cholesterol as noted in the model membrane experiments. The results of growth inhibition are shown in Table 4 and show that the cholesterol producing strain was 16-fold less sensitive toward natamycin compared with the corresponding wild type. This demonstrates that also *in vivo* natamycin has a strong specificity for ergosterol over cholesterol. Moreover, given the difference in chemical structures of ergosterol and cholesterol, the importance of the double bonds of the B-ring for interaction with natamycin is further emphasized consistent with the results of the *erg* strains. Nystatin had the same effect on the yeast strains as natamycin, whereas filipin is apparently less specific as it was almost as effective in killing the cholesterol producing strain as the wild type strain.

To determine whether the inhibition of growth was related to the amount of binding of natamycin to these yeast strains, a binding assay with the different strains was performed. All the strains were tested and in addition an *E. coli* wild type strain was taken as a negative control, because it contains no sterols in the plasma membrane. For clarity only 6 strains are depicted in Fig. 5A. The highest amount of binding of natamycin was observed for the wild type (both strain RH448 and RH6611), together with *erg4Δerg5Δ*. The least amount of binding was observed for the negative control, the *E. coli* wild type strain, whereas strain *erg2Δerg6Δ* showed the least amount of binding of the yeast strains. The relation of the amount of binding of natamycin to the MIC values is depicted in Fig. 5B, at a cell density corresponding to an A₆₀₀ of 10. The figure shows an inverse relation between the amount of bound natamycin to the MIC value of a particular strain, strongly suggesting that the differences in MIC value toward natamycin are directly related to the difference in binding of natamycin to the yeast cells. In addition, binding studies with vesicles made from lipid extracts of plasma membrane-enriched yeast membrane fractions were per-

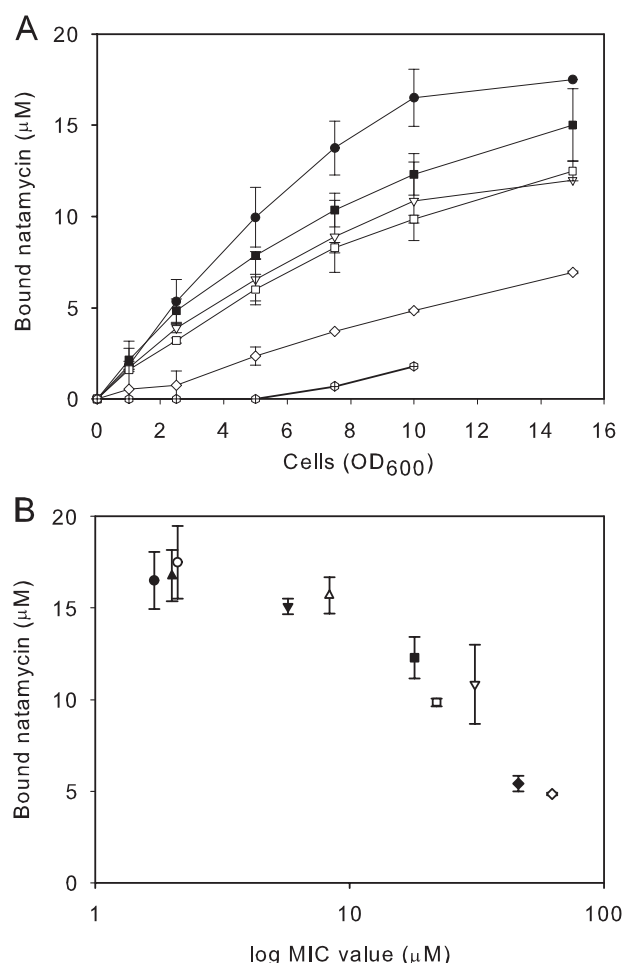


FIGURE 5. Binding of natamycin to different yeast strains. In A, the binding of natamycin with the yeast strains is depicted by addition of 30 μ M natamycin to varying cell densities (A_{600}). In B the binding of natamycin to yeast at an A_{600} of 10 is plotted against the MIC values of the different strains. The binding was determined in duplicate in 10 mM MES/Tris, 15 mM K_2SO_4 , pH 7.0, and the strains examined were the wild type RH448 (●), the wild type RH6611 (▲), erg4Δerg5Δ (○), erg3Δ (▼), erg6Δ (△), erg3Δerg6Δ (■), erg2Δ (□), cholesterol (▽), erg2Δerg3Δ (◆), erg2Δerg6Δ (◇) and the *E. coli* wild type strain (197).

formed and resulted in a similar binding pattern as compared with intact yeast cells (data not shown).

Effect of Polyene Antibiotics on Proton Permeability in Vitro—The binding assays as well as the MIC determinations show that there is a specific interaction of natamycin with ergosterol, which leads to an inhibition of cell growth. To test if the interaction of natamycin with ergosterol leads to changes in membrane permeability, different leakage assays were employed. Natamycin did not produce any carboxyfluorescein leakage from DOPC vesicles containing 10 mol % ergosterol in contrast to filipin, which did cause carboxyfluorescein leakage (results not shown). Because nystatin, which is known to form pores, also did not cause carboxyfluorescein release from the vesicles, the pores formed by this antibiotic are apparently too small to allow passage of this dye. A similar situation could be the case for natamycin. Therefore, we tried an assay based on leakage of protons that should be small enough to pass such pores. This assay makes use of a pH-dependent fluorescent probe (HPTS), which has a high fluorescent intensity at neutral pH and a low

fluorescent intensity at low pH (9). An example of the effect of 5 μ M polyene antibiotics on 10% ergosterol containing vesicles is given in Fig. 6A. Trace 1 was recorded by addition of the vesicles to the cuvette and following the fluorescence intensity in time (the blank). After $\sim$ 300 s, the detergent DDAO was added to dissipate the vesicles and the fluorescent intensity reaches its lowest point. Nigericin (trace 2) was used as a positive control and resulted in an immediate dissipation of the proton gradient over the model membrane. Indeed, filipin (trace 3) and nystatin (trace 4) both resulted in leakage of the membrane vesicles. Strikingly, natamycin (trace 5) did not result in proton leakage at this concentration. A more quantitative analysis of the effect of the antibiotics on H^+ leakage in model membranes is given in Fig. 6, B–D.

The results show that in strong contrast to filipin and nystatin, natamycin did not induce any significant proton leakage in ergosterol containing vesicles even at very high concentrations (Fig. 6D). This would indicate that natamycin does not act via a perturbation of the membrane barrier and thus has a completely different mode of action compared with filipin or nystatin. To test if similar effects could be observed *in vivo*, a proton leakage assay in yeast was performed.

Effect of Polyene Antibiotics on Proton Permeability and Growth in Vivo—To correlate the results from the *in vitro* leakage assay to an *in vivo* effect, yeast cells were loaded with the pH-sensitive probe CFDA-SE. The effect of the polyene antibiotics added at 2-fold the MIC value on the wild type yeast strain is displayed in Fig. 7. The fluorescence of the loaded yeast strain was monitored after different time intervals (Fig. 7A). In the absence of antibiotic, the yeast cells displayed a steady fluorescence intensity that decreased slightly in time. When natamycin was added, no further decrease in fluorescence intensity was observed. When nystatin was added to the yeast cells an immediate decrease in fluorescence intensity was observed, most likely due to the formation of pores in the plasma membrane. After the decrease of fluorescence, a gradual increase of the fluorescence intensity was observed, which indicates that the yeast cells try to restore the ion gradient over the plasma membrane. Fig. 7B shows that with the same conditions used to study the antibiotic-induced release of protons, growth was inhibited by both natamycin and nystatin further emphasizing the difference in mode of action between these polyene antibiotics. In conclusion, natamycin does not kill yeast cells by permeabilizing the plasma membrane.

DISCUSSION

In this study we have demonstrated that natamycin kills yeast by specifically binding to ergosterol but without permeabilizing the plasma membrane. This novel mechanism sets natamycin apart from other polyene antibiotics studied so far. We included two of these as a reference in this study.

The ITC and direct binding studies in both model and yeast membrane systems demonstrated that natamycin binds with an apparent affinity of $\sim$ 100 μ M specifically to ergosterol with a stoichiometry of $\sim$ 1:1 or 1:2 depending upon whether the sterol is available for interaction only in the outer leaflet or in both leaflets of the membrane. This stoichiometry range is in good agreement with the stoichiometry reported before for other

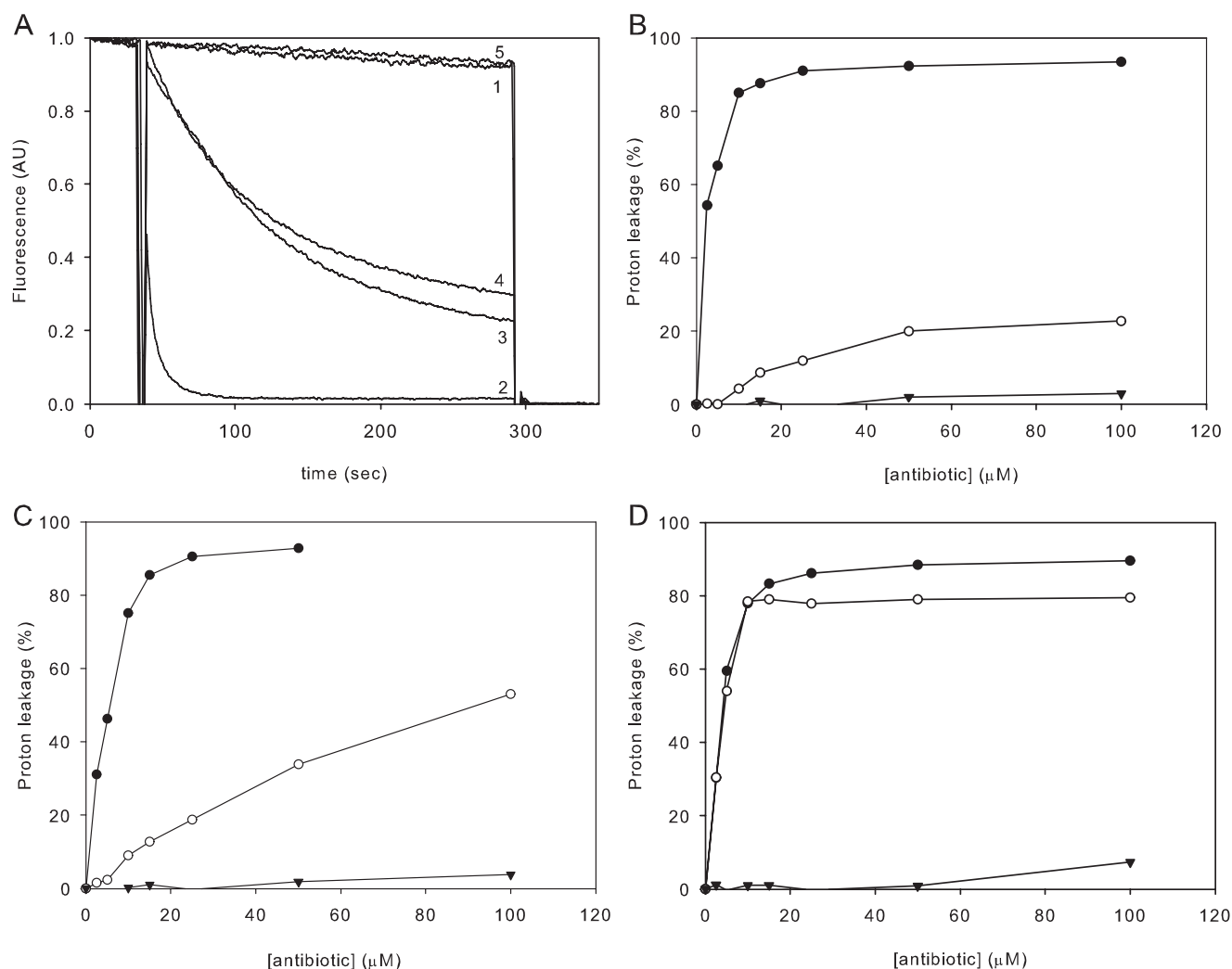


FIGURE 6. Effect of the polyene antibiotics on the proton permeability of membrane vesicles. A, time courses of HPTS fluorescence, which was influenced by: 1) no addition, or the addition of: 2) nigericin; 3) filipin; 4) nystatin; and 5) natamycin ($5 \mu\text{M}$ antibiotic) to 10% ergosterol containing vesicles. B–D, the percentage of proton leakage was determined by adding various concentrations of filipin ($\bullet$), nystatin ($\circ$), and natamycin ($\blacktriangledown$) to vesicles containing (B) no sterols, (C) 10% cholesterol, or (D) 10% ergosterol. Measurements were performed in 10 mM MES, 0.2 M Na_2SO_4 buffer, pH 5.5, and the vesicles had a $35 \mu\text{M}$ final phospholipid concentration.

polyene antibiotic-sterol interactions (14). However, given the complexity of the binding data and the unknown nature of the natamycin-ergosterol complex a more quantitative discussion of the binding data is not possible.

Both results from the model system and the yeast mutants gave a clear picture of the requirements within the sterol structure for the binding to natamycin, where only variations in the double bonds of the B-ring resulted in large differences in interaction, especially the sp^2 hybridization of C-7. The packing of the sterol molecule together with natamycin is probably related to this structural requirement. The conformation of ring B in ergosterol differs from the conformation of this ring in cholesterol, which is clearly illustrated in Fig. 8. The sp^2 hybridization at C-7 in ergosterol (indicated with an arrow, Fig. 8A) results in a 1,3-diplanar chair conformation, which is lacking in cholesterol giving a half-chair conformation (Fig. 8B) (15). Natamycin has a tightly constrained molecular topology that gives a very high apparent structural order (16). Therefore it is very likely that the diplanar chair conformation of the B-ring in ergosterol will result in a more efficient interaction. For amphotericin B,

similar results were observed, where the sp^2 hybridization at C-7 was of critical importance for the interaction of this antibiotic with sterols in model membranes, whereas the double bond at C-5,6 was not essential (17).

The sterol specificity of natamycin in model and biomembranes was more comparable with nystatin than to filipin. This can also be observed from the additional ITC experiments that are given as supplemental data. The observed order of binding for filipin in the ITC experiment was 10% ergosterol > 10% cholesterol > 0% sterol leading to the values of 41.3, 20.4, and $17.4 \times 10^4 \text{ M}^{-1}$, respectively. Filipin did not seem to be as dependent on sterol structure nor the presence of sterols as the apparent K values to different membranes did not vary much (in agreement with literature) (18–20). The binding of nystatin seemed more similar to natamycin and the K value is slightly lower compared with natamycin, 2.72 to $5.7 \times 10^4 \text{ M}^{-1}$.

We have shown that the interaction between natamycin and ergosterol leads to an inhibition of yeast growth and cell death, but, this is not via a permeabilization of the membrane as is exhibited by nystatin. The structure of the natamycin-ergos-

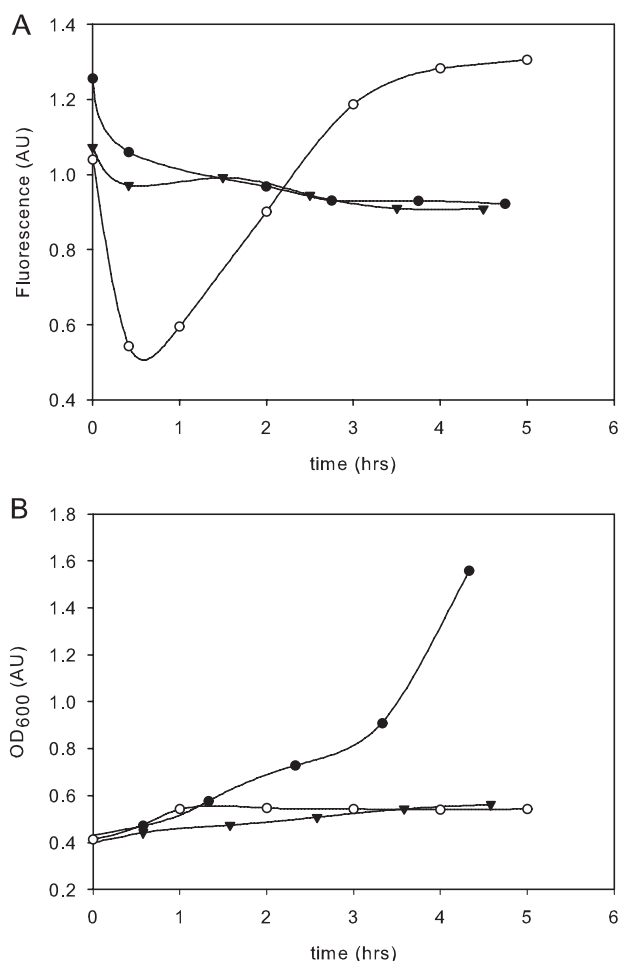


FIGURE 7. Effect of the polyene antibiotics on CFDA-SE loaded wild type yeast cells. Yeast cells in YPUADT medium buffered with 50 mM citric/phosphate (pH 4) were followed in time after no addition of antibiotic (●) or the addition of nystatin (○) or natamycin (▼) (2.5 μ M). **A**, the fluorescent intensities; and **B**, the optical densities were monitored at regular time intervals.

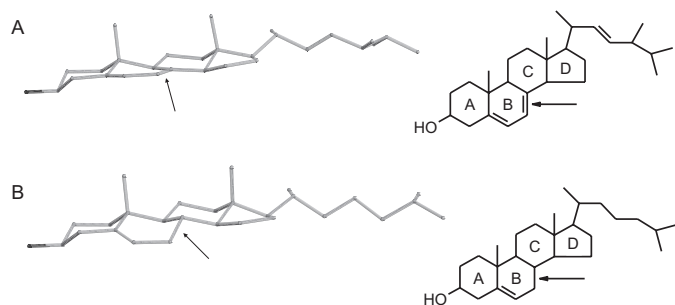


FIGURE 8. The conformation of the ring structures in ergosterol (A) and cholesterol (B) viewed from the side at approximately the same angle, together with the flat structures. The arrow indicates the C-7,8 bond, resulting in different B-ring conformations; a 1,3-diplanar chair conformation in ergosterol and a half-chair in cholesterol. Structures were taken from crystal structures given in Refs. 28 and 29.

terol complex is unknown, but assuming that it is similar to nystatin-ergosterol complexes, two possible explanations can account for the difference in mode of action. One is that the formed complex of natamycin and ergosterol might be too tight to pass even an ion as small as a proton. Second, the formed complex could be too small to span the complete bilayer. If the mode of action of natamycin does not involve permeabilization,

then how does it act? In this light it is worth recalling that for the polyene antibiotics that are known to permeabilize the membrane, also other modes of actions have been proposed such as oxidative damage of membrane structures (21–23). The mode of action of natamycin must be related to an important function of ergosterol in the yeast cells. For example, sterols are known to have an ordering effect on the membrane, it is thought that they reside in specific sterol-rich domains in membranes and they are also known to be involved in endocytosis, exocytosis, and vacuolar fusion (24–27). Natamycin might inhibit these important processes by binding to ergosterol such that the sterol cannot perform its functional effects.

Acknowledgments—We thank M. R. van Leeuwen and J. Dijksterhuis (Fungal Biodiversity Center (CBS), The Netherlands) for helpful comments and valuable research discussions.

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Natamycin Blocks Fungal Growth by Binding Specifically to Ergosterol without Permeabilizing the Membrane

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J. Biol. Chem. 2008, 283:6393-6401.

doi: 10.1074/jbc.M707821200 originally published online December 29, 2007

Access the most updated version of this article at doi: [10.1074/jbc.M707821200](https://doi.org/10.1074/jbc.M707821200)

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